Modeling and Simulation Credibility Assessment of Explicit FEA for Total Knee Replacement Wear and Contact Mechanics

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Abstract

Objective. This study assesses the credibility of two explicit finite element analysis (FEA) models for predicting polyethylene wear, contact mechanics, and backside interface micromotion in a posterior-stabilized total knee replacement under mechanism-specific conditions. One model resolves component-level interactions under kinematics-driven inputs, and the other couples components with anatomy-inspired constraints under load-driven inputs. Approach. We defined four clinically relevant mechanisms—steady-state tibial insert wear during level walking, varus–valgus edge loading, backside interface micromotion and fretting, and high-flexion pivoting cross-shear—and simulated them with each model. A cross-shear–dependent Archard formulation was used for wear. We implemented contact via an explicit penalty approach with single-surface search, hourglass control for reduced integration, and selective mass scaling as needed to maintain stability. Quantities of interest included peak and mean contact pressures, contact area, sliding distance, cross-shear metrics, wear scar evolution and volumetric loss, and micro-slip at the insert–tray interface. Evidence. Physical test data came from three independent, GLP-documented simulator campaigns and benchtop measurements of materials and locking features for the same implant family. We compared simulations with level-walking simulator wear volume and contact patch maps, pin-on-plate cross-shear tests, and tibial insert–tray micro-motion rigs. We implemented structured unit tests, manufactured solutions, and benchmark contact cases to shore up numerical credibility, and carried out sensitivity and uncertainty quantification across loads, friction, and material parameters. A private, continuous 0–12 index was used to score each model–mechanism pair across eight factors: lineage of the evidence, suitability of the chosen governing equations and constitutive laws, implementation tests, solution quality, agreement with measured outcomes, stability of results to plausible alternatives, characterization of residual uncertainty, and vulnerability to misuse. Findings. For gait wear, Model 1 reproduced simulator volume loss within 12 percent RMS after calibrating a single global wear coefficient, and cross-shear scaling captured the observed medial–lateral asymmetry. Edge loading increased peak pressure above 100 MPa and shifted the wear scar toward the rim; high-flexion pivoting increased predicted wear rate by 1.5–1.8× relative to gait. Backside micromotion stayed below 15 microns for nominal locking but rose above 40 microns under tray misfit. In Model 2, ligament constraints moderated posterior roll-back and limited tibial insert rim contact during varus–valgus maneuvers; nonetheless, under 5 degrees of valgus and 2 mm medial lift-off, peak pressures exceeded 110 MPa. Global Sobol analysis identified friction and cross-shear exponent as the dominant contributors to wear prediction variance. Credibility. Across the eight model–mechanism pairs, median scores exceeded the developing threshold and approached established for gait wear. Priorities to elevate credibility include richer edge-loading experiments, direct high-flexion pivoting wear data, and expanded characterization of backside locking compliance. The presented scoring and evidence trail support using the models for comparative design assessment and what-if studies, with caution for extreme postures and tray-insert tolerances.

Keywords: finite element analysis, total knee replacement, wear modeling

1. Introduction

Ultra-high-molecular-weight polyethylene (UHMWPE) wear remains a central determinant of total knee replacement (TKR) durability. Posterior-stabilized designs, with a cobalt-chromium femoral component articulating against a polyethylene tibial insert fixed to a modular tray, experience complex combinations of rolling, sliding, lift-off, and pivoting across activities. Volumetric loss and wear scar evolution arising from these motions can lead to elevated debris generation, osteolysis, and, in some cases, accelerated insert damage when edge loading occurs. With the proliferation of computational tools, explicit finite element analysis (FEA) has emerged as a practical option to resolve nonlinear, transient contact under anatomically relevant motions and to accumulate wear over millions of cycles with acceptable turnaround times. Yet, model credibility in this setting depends on structured evidence that the computational representations are suitable for their intended comparisons and decisions.

This paper assesses the credibility of two explicit FEA models for a single posterior-stabilized TKR family across four mechanisms known to stress wear and contact behavior.

The first model focuses on component-level interactions under kinematics prescribed to the femoral component, synthesizing simulator-style inputs to drive relative motion with no surrounding soft tissues. The second couples the articular components with ligament-inspired constraints under load-driven conditions, allowing malalignment, lift-off, and high-flexion postures to emerge from applied forces and torques. For both, we quantify the degree to which the predicted contact mechanics, cross-shear, and wear comport with physical measurements, and we score the evidence using a continuous index that recognizes that credibility is not binary but graduated across use contexts.

Because laboratory test setups vary, we refer to apparatus details where relevant but avoid harmonizing all external protocols; instead, we explicitly engage with the variations and define mapping steps into model inputs. The historical pathways by which implants accrue usage in vivo also differ markedly by patient and region; this work confines itself to the boundary cases that can be induced in a bench setting and does not attempt to reconstruct long-term patient-level exposure patterns.

2. Device Description and Materials

The device is a posterior-stabilized TKR comprising a cobalt-chromium-molybdenum (CoCrMo) femoral component, a UHMWPE tibial insert mated via a central locking feature to a titanium alloy modular tray, and a keeled stem. The femoral component has medial and lateral condylar radii transitioning from 21.5 to 18.3 mm in the distal-to-posterior wrap, with a sagittal J-curve that defines roll-back. The tibial insert articulating surfaces are conforming in the medial compartment and slightly less conforming laterally, generating coupled rolling and sliding during flexion. The post-cam mechanism engages beyond approximately 60 degrees of flexion to limit posterior translation of the femoral component.

Material properties were established from specimen-specific measurements when possible. UHMWPE tensile tests (ASTM D638 Type IV) furnished an elastic modulus of 0.85–1.05 GPa at room temperature with a 0.35–0.40 Poisson ratio, and uniaxial compression creep tests informed a Kelvin–Voigt element with a 1.0 GPa spring and an effective dashpot of 0.25 GPa·s for short-time settling. CoCrMo was modeled as linear elastic with E = 210 GPa, ν = 0.30; titanium alloy tray was E = 110 GPa, ν = 0.33. The locking mechanism engages via a taper and snap-fit features; pull-off tests on five inserts yielded an axial retention force of 1.8–2.4 kN and lateral compliance of 0.02–0.05 mm/kN under medial–lateral shear.

The experimental wear dataset supporting this work originated from three independent simulator campaigns conducted under GLP with the same implant family and traceable lot numbers for the inserts; the polyethylene characterization, including crosslink density and oxidation indices, accompanies each lot's records.

3. Computational Models and Explicit FEA Workflows

We developed two explicit FEA models. Model 1 (Component-Level Explicit FE) resolves the femoral component, polyethylene insert, and modular tibial tray under prescribed six-degree-of-freedom kinematics of the femoral component relative to the tray, including flexion-extension, axial rotation, anteroposterior translation, and varus–valgus angulation. This kinematics-driven approach matches standard wear simulator inputs, providing controlled, repeatable relative motion at the interface. No bones or musculotendon structures are included. Model 2 (Anatomy-Coupled Explicit FE) augments the articular construct with nonlinear spring sets representing the medial and lateral collateral ligaments, the posterior cruciate substitute function via post-cam, and anterior soft-tissue constraints; it is driven by applied compressive loads, torques, and prescribed motions representative of whole-joint boundary conditions. The ligament springs use tension-only, piecewise-exponential force–elongation laws calibrated from cadaveric data, with slack lengths tuned to nominal knee laxity envelopes.

Both models use hexahedral-dominant meshes for the polyethylene with local refinement in expected high-contact regions; a typical insert mesh has 120,000–160,000 elements with 0.5–0.8 mm characteristic length in the contact zone and 1.5–2.0 mm in the bulk. The femoral component and tray are modeled with rigid surfaces for contact to reduce computational cost; additional flexibility of the tray is handled via a thin elastic layer representing the tray bed under the insert in Model 2. Surface contact is implemented via a penalty-based algorithm with a single-surface contact search on the polyethylene and master surfaces on the metal, with segment-to-segment enforcement to mitigate chatter. A Coulomb friction law with a nominal coefficient between 0.03 and 0.06 is used; in sensitivity studies we also used a velocity-weakening friction curve to represent lubrication effects under different sliding speeds.

Wear is computed incrementally using an Archard-type formulation extended to account for cross-shear. Specifically, an effective wear coefficient k_eff = k0·f(CS) is applied, where CS is a scalar cross-shear measure between 0 and 1 computed from the principal direction changes of sliding at the surface; f(CS) = 1 + α·CS^β, with α and β calibrated to pin-on-plate tests under matched polyethylene and CoCrMo surface finishes. Local wear depth updates via Δh = k_eff·p·Δs/H, where p is contact pressure, Δs the sliding increment, and H the hardness surrogate (taken as 0.9 GPa for UHMWPE under the tested conditions). Creep is represented by a Kelvin–Voigt viscoelastic layer superposed on the elastic response; the accumulated creep strain is reset per million-cycle increment to match dwell-time distributions in the simulator campaigns.

To provide implementation assurance, we built 38 small test problems for component algorithms: contact pair registration and normal/tangent vector alignment; penalty stiffness update with variable time step; sliding distance accumulation and cross-shear vector update; and wear depth projection to the current surface. Two manufactured-solution checks—one for a sinusoidal sliding over a flat elastic foundation and one for a Hertzian line contact with analytical pressure distribution—returned L2 norm errors below 1.0e-6 in the displacement and pressure fields, respectively, when run with double precision and tightened penalty enforcement.

We employed an explicit central-difference integrator with selective mass scaling to maintain a minimum stable time step between 0.10 and 0.15 microseconds for the refined insert meshes without violating inertial neglect. Hourglass control used a stiffness-based scheme tuned to maintain the nonphysical energy fraction under 0.5 percent. Across all runs reported here, the ratio of kinetic to internal energy stayed under 2 percent after the initial ramp; repeated simulations at half the time step changed peak contact pressure by less than 0.8 percent and wear volume per 100,000 cycles by less than 1.2 percent.

4. Mechanisms Assessed: Definitions and Performance Metrics

We examined four mechanisms of interest, each associated with clinical observations and laboratory tests. First, steady-state tibial insert wear during level walking reflects long-duration service under relatively repeatable motion and load cycles. The simulator-style waveform we reference consists of a 0.5–3.0 kN compressive load, 0–58 degrees flexion, ±6 degrees internal–external rotation, and 7–11 mm anteroposterior translation over a 1-second cycle. Second, varus–valgus edge loading occurs when malalignment or soft-tissue imbalance shifts load toward the rim, reducing conformity and concentrating contact stress. We operationalize this by superimposing 3–5 degrees of varus or valgus and 1–2 mm compartmental lift-off. Third, backside interface micromotion and fretting can produce local damage and debris at the underside of the insert; we analyze relative motion between the insert and tray bed, especially in response to shear and torque inputs. Fourth, high-flexion pivoting cross-shear is characterized by combined axial load with transverse pivoting of the femur on the tibia near 80–110 degrees of flexion; this condition can elevate cross-shear beyond gait-only conditions and has been implicated in accelerated wear under certain activities.

For each mechanism, we define a set of quantities of interest. Contact mechanics include peak and mean contact pressures in each compartment, contact area, and locations of peak stress. Wear-related outputs include wear depth distributions, volumetric loss per million cycles, and wear scar centroid shifts. Cross-shear is quantified by a scalar derived from changes in the sliding direction; we report its time-averaged value per element weighted by contact work. Backside micromotion is captured by relative displacements at nodal sets on the insert underside relative to the tray, and we report maximum and 95th-percentile micro-slip amplitude during cyclic shear loading.

5. Loads, Boundary Conditions, and Contact/Wear Formulations

Model 1 uses a kinematics-driven load case matching an ISO-style simulator for level walking, with five additional variants to interrogate mechanism-specific behavior: varus and valgus angulation of 3 and 5 degrees combined with 1 and 2 mm of medial or lateral lift-off, a high-flexion pivoting series at 90 and 110 degrees with ±6 Nm axial torque and 2.0–3.0 kN compressive load, and a backside shear series with 300–600 N lateral force applied at the insert's articular surface while constraining the tray to ground. The femoral component's motions are prescribed as smoothed time histories to avoid start-up transients, and 10 cycles are run to condition the contact before wear accumulation begins. Wear is accumulated over blocks of 100,000 cycles per analysis increment and mapped back to the surface using shape functions to update the geometry.

Model 2 employs load-driven conditions. Compressive loads rise from 0.6 to 3.2 kN per cycle for gait, and internal–external torques of ±4 Nm with anteroposterior shear up to ±150 N are applied to the femur. Ligament springs engage based on relative displacements between femoral and tibial attachment points, with medial collateral slack set at 0.5 mm and lateral at 0.8 mm in the nominal case. Posterior constraint arises either from post-cam engagement in deep flexion or from posterior spring elements beyond 70 degrees. Malalignment is introduced by rotating the tibial tray about the mechanical axis with preloads held constant, allowing the anatomy-coupled system to settle into a new equilibrium. For backside micromotion, the insert's locking taper is modeled with contact and a weak axial compliance calibrated to pull-off tests; micro-slip under alternating shear is recorded.

The wear update employs the same cross-shear–sensitive Archard law in both models. We calibrated $\alpha = 0.9$ and $\beta = 0.7$ to reproduce a 1.7× increase in wear rate observed between unidirectional and 90-degree reciprocating pin-on-plate tests at a nominal k0 of 3.5e-7 mm3/N·m. We further tuned the effective hardness surrogate to reproduce the onset of material removal under mixed rolling–sliding observed in component-level tests. Contact penalty stiffness was selected so that average penetration at peak load remained under 0.5 percent of the local element size. Normal/tangential coupling used a predictor–corrector within each explicit step to limit slip overshoot. The friction coefficient baseline was 0.04 for wet, serum-lubricated surfaces, with a tested range of 0.02–0.08 for sensitivity and uncertainty propagation.

Because solution quality in explicit dynamics can degrade under coarse time discretization, we constrained the minimum time step via element characteristic length and wave speed. In all runs after the initial ramp-in, the internal energy dominated the kinetic energy; the impulse metric normalized by contact duration—monitoring spurious oscillations—remained flat across cycle count. These controls ensured that the local power density driving wear updates was not artifactual.

6. Credibility Assessment Methods and Scoring Rubric

We scored credibility for each model–mechanism pair using a private continuous index from 0 to 12, termed the Explicit FEA Credibility Index. The interpretive thresholds at 3, 6, and 9 demarcate screening, developing, and established evidence levels, respectively. Scores are assigned independently for ten topics organized as eight clear elements and two that are contextually uncertain. The eight clear elements encompass lineage and traceability of supporting information, suitability of the governing equations and constitutive relations to the phenomena, checks that the implemented algorithms match

Figure 1: Schematic of the two explicit FEA workflows. Model 1 applies simulator-style kinematics to the femoral component over a meshed polyethylene insert seated in a tibial tray; Model 2 replaces the kinematics prescription with load-driven constraints and nonlinear ligament springs while retaining the same contact and wear formulations.

their specifications, controls for discretization and integration quality, agreement with measured system outputs, stability of conclusions to plausible changes in inputs and numerics, quantitative accounting for residual spread in outputs, and vulnerability to misuse in setup or interpretation. The two ambiguous topics, test conditions and use history, are scored to reflect available context yet are not as readily pinned down for this device and activity space.

We adopted a uniform scoring procedure across all pairs. For each topic, we defined a rubric that anchors 0 as absent or inapplicable evidence and 12 as comprehensive, multi-source, quantitatively bounded support. Raters considered: number and independence of data sources; proximity of test articles and conditions to the model instantiation; explicit physics captured and omitted; degree of implementation checking; indicators of numerical quality and convergence; number and breadth of output comparisons; parametric stress tests and alternative formulations tested; breadth and execution of uncertainty quantification; and existence of safeguards and documented usage patterns. Where applicable, we assigned half-point increments to reflect gradations.

Scoring was performed by two analysts independently; disagreements larger than 1 point were reconciled through joint review of the evidence dossier. All inputs to scoring—trace matrices from data to parameters, descriptions of approximations and their rationale, test summaries, and numerical quality metrics—are cataloged and versioned. The per-pair matrices are reproduced in the supplemental section to support reuse and external review.

Data pedigree — gradation.

a.
No documented provenance; untraceable or synthetic information used without justification.

b.
Single-source measurements with partial traceability and limited relevance to the modeled device or conditions.

c.
Multiple independent sources with traceable specimens and conditions mapped to the model; gaps documented.

d.
Comprehensive, independent sources with GLP or equivalent documentation, specimen-level traceability, and direct mapping to model parameters and scenarios.

7. Results: Model 1 Across Four Mechanisms

Steady-state tibial insert wear during level walking in Model 1 produced medial-lateral asymmetry consistent with simulator observations. Under the nominal friction coefficient of 0.04, peak medial contact pressure was 62–68 MPa, lateral peak 48–55 MPa, with contact areas of 350–420 mm2 medially and 270–320 mm2 laterally over mid-stance. The time-averaged cross-shear scalar was 0.23 on the lateral side and 0.12 on the medial side, leading to a higher effective wear factor laterally despite lower pressure. Predicted volumetric loss per million cycles was 25–31 mm3 across the tested inserts after calibrating a single global wear coefficient k0 to match the mean of the three GLP campaigns for the same size. Against six million-cycle datasets with measured gravimetric loss and reconstructed wear scar maps, the RMS difference in wear volume after calibration was 12 percent, and the Jaccard index for the projected wear scar footprint exceeded 0.72, indicating appropriate localization of wear. The predicted post-cam engagement altered posterior contact distribution in late stance but contributed less than 5 percent to the total wear increment.

Varus–valgus edge loading was recreated by imposing 3 and 5 degrees of malalignment with 1 or 2 mm lift-off. Under 5 degrees valgus and 2 mm lateral lift-off, the contact patch migrated to the lateral rim, producing a peak pressure of 105–112 MPa and reducing the effective contact area by 38 percent compared with nominal gait. The wear scar centroid shifted 3.4–4.2 mm laterally, and the local cross-shear rose to 0.31, with a 1.6× increase in local wear factor; yet the volumetric wear increment per million cycles increased only 18–24 percent due to the reduced area of contact and shorter sliding distance in the destabilized configuration. Under 3 degrees varus with 1 mm medial lift-off, the medial side exhibited 96–101 MPa peak pressure, with similar qualitative changes. These results align with reported sensitivity of pressure and wear to alignment, showing marked local intensification without equivalently large integral wear increases.

Backside interface micromotion was evaluated by applying lateral shear of 300–600 N under a 1.5 kN compressive bias. With nominal locking engagement matching the measured taper fit and compliance, maximum micro-slip amplitude at the insert–tray interface was 9–14 microns at 300 N and 12–17 microns at 600 N, with 95th-percentile values 5–8 microns lower than the maxima. Introducing a 0.05 mm axial gap (representing wear-in or manufacturing tolerance exceedance) increased the maxima to 28–42 microns at 600 N, while a reversed taper angle case (a deliberately incorrect assembly case) produced ratcheting up to 65 microns. The distribution of micro-slip was concentrated near the anterior and posterior edges under alternating shear. These values correspond to fretting regimes reported in literature, suggesting potential for debris if such gaps persist.

High-flexion pivoting cross-shear was simulated with 90 and 110 degrees flexion under 2.5–3.0 kN compressive loads and ±6 Nm axial torque. During pivoting, the sliding direction rotated approximately 50–70 degrees over a half-cycle, yielding time-averaged cross-shear scalars of 0.35–0.42 depending on torque amplitude. Peak contact pressures were 58–64 MPa on the pivoting side at 90 degrees flexion, rising to 72–78 MPa at 110 degrees due to increased posterior wrap and reduced conformity. The combined effect of elevated cross-shear and comparable pressures increased the predicted wear rate by 1.5–1.8× relative to level walking on the pivoting side. The non-pivoting side saw minimal change in wear relative to gait.

8. Results: Model 2 Across Four Mechanisms

Level walking in Model 2, driven by compressive loads and torques with ligament constraints, produced contact mechanics similar to Model 1's nominal gait but with moderated lateral rotation and posterior roll-back due to collateral engagement. Peak medial pressure was 60–66 MPa, lateral 46–52 MPa, and the time-averaged cross-shear was modestly lower (0.10 medial, 0.21 lateral). The volumetric wear per million cycles was 24–29 mm3 using the same global k0, with wear scar footprints nearly identical to Model 1. Small differences in scar orientation arose from the ligaments distributing rotation more evenly. At the post-cam threshold, Model 2 exhibited a gradual rather than abrupt transition in posterior contact thanks to spring engagement, reducing transient pressure spikes.

Varus–valgus edge loading in Model 2 demonstrated the moderating influence of ligamentous constraints. Under applied valgus rotation of 5 degrees with maintained compressive load, the lateral collateral slack was exhausted, and the medial spring elongated, resisting full lift-off. A remaining 1.2 mm lateral compartment separation nonetheless occurred under peak load, during which peak lateral pressure exceeded 110 MPa and effective contact area dropped by 41 percent. The ligaments mitigated further rim excursion compared to Model 1, yet the magnitude of peak contact still entered the range associated with plastic risk in polyethylene. Wear increments per million cycles rose by 14–19 percent relative to gait; the reduction relative to Model 1's lift-off case reflects the dynamic redistribution enacted by the springs.

Backside interface micromotion in Model 2 mirrored Model 1's nominal locking case under shear, with maxima of 10–16 microns at 300–600 N shear. When valgus malalignment was present, the posterior-lateral region experienced higher micro-slip, with maxima 20–26 microns under 600 N. Imposing a 0.05 mm axial gap again increased micro-slip by a factor of roughly 2, indicating that tolerance excursions dominate interface motion more so than soft-tissue constraints do. The anatomy-coupled model's inclusion of a thin elastic tray bed layer introduced slightly larger distributed motion across the bed relative to a rigid tray, raising 95th-percentile micro-slip by 2–4 microns but keeping maxima within the same bands.

High-flexion pivoting cross-shear in Model 2 was examined under 90 and 110 degrees flexion with ±6 Nm torque and 2.8–3.2 kN compressive load. The ligament constraints and post-cam mechanism combined to limit the extent of transverse pivot at 90 degrees, resulting in cross-shear scalars of 0.31–0.37. At 110 degrees, pivoting increased, yielding 0.38–0.43. Peak contact pressure rose to 75–82 MPa on the pivoting side at 110 degrees, and the predicted wear rate increased by 1.4–1.7× compared with gait. Compared with Model 1, Model 2's pivoting cross-shear values were slightly lower at 90 degrees due to ligament restraint but converged at 110 degrees where post-cam engagement dominated.

9. Credibility Scoring by Model–Mechanism Pair: Data Pedigree

We assembled specimen-level links between each model parameter and its source measurements, and we mapped simulator campaigns to the modeled implants and sizes. Independent datasets for wear and micromotion, as well as benchtop material characterizations, supported cross-checks. The supplemental matrices provide the detailed mapping and the assigned index for each pair based on the breadth, independence, and directness of the sources.

Data pedigree — gradation.

a.
Inputs and comparisons rely on anecdotal or untraceable sources.

b.
Some traceable data exist, but key parameters or comparisons derive from indirect sources.

c.
Multiple traceable datasets underpin parameters and outputs; minor gaps are documented.

d.
Comprehensive, independent, and directly relevant datasets with full traceability support all parameters and outputs.

10. Credibility Scoring by Model–Mechanism Pair: Model Form

We evaluated the governing equations and constitutive choices relative to each mechanism. The explicit contact and cross-shear–dependent wear law align with short-time-scale contact and long-time-scale accumulation. Choices include linear elasticity for the metals and a viscoelastic representation for UHMWPE creep, with plasticity not invoked due to predicted pressure magnitudes below yield in most scenarios. Omitted phenomena—porosity evolution, oxidative degradation, and third-body abrasion—are documented, and the proximity of

operating conditions to regimes where such effects matter was considered during scoring. The supplemental matrices list the pair-specific judgments.
Model form — gradation.

a.
Core physics missing or inappropriate; assumptions invalidate conclusions.

b.
Essential physics present but important mechanisms are simplified without justification.

c.
Governing equations and constitutive laws match the phenomena; simplifications are justified and bounded.

d.
High-fidelity physics specific to mechanisms are included; residual simplifications are shown to be immaterial to decisions.

11. Credibility Scoring by Model–Mechanism Pair: Numerical Code Verification

Implementation checks included unit tests for contact search and enforcement, sliding distance accumulation, and wear update, as well as manufactured solutions and benchmark problems for pressure distributions. These checks were run at each software revision used to produce the results herein. The pair-wise scores reflect the relevance of these tests to the contact geometries and loading paths of each mechanism.
Numerical code verification — gradation.

a.
No targeted tests of implementation correctness exist.

b.
Limited checks of select modules; coverage gaps remain for key algorithms.

c.
Systematic unit tests and benchmark comparisons cover core algorithms relevant to the application.

d.
Extensive testing including manufactured solutions, regression tests, and independent code reviews cover all critical algorithms.

12. Credibility Scoring by Model–Mechanism Pair: Numerical Solver Error

We assessed discretization and integration quality via mesh refinement in contact regions, step-size controls for explicit integration, and energy metrics. Checks included halving the time step and local mesh sizes, monitoring kinetic-to-internal energy ratios, and ensuring hourglass energy remained a small fraction of the total. The assigned indices account for evidence completeness under each mechanism's loading path.
Numerical solver error — gradation.

a.
No evidence of numerical quality; discretization and time-step effects are unknown.

b.
Spot checks exist but do not directly address the most demanding scenarios.

c.
Targeted studies show stable solutions and small changes upon refinement for relevant scenarios.

d.
Comprehensive convergence and stability demonstrations across all scenarios of interest with quantitative tolerances met.

$$\text{CI\_\{score\}} = \frac{\sum_{i=1}^{n} w_i \, x_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

$$\text{\textbackslash varepsilon\_\{rel\}} = \frac{\sum_{i=1}^{n} w_i \, x_i}{\sum_{i=1}^{n} w_i} \tag{2}$$

$$\text{U\_\{val\}} = \frac{\sum_{i=1}^{n} w_i \, x_i}{\sum_{i=1}^{n} w_i} \tag{3}$$

13. Credibility Scoring by Model–Mechanism Pair: Output Comparison

Comparisons include wear volumes, wear scar morphologies, contact patches, and backside micromotion amplitudes against independent measurements where available. For gait wear, the comparisons leverage multi-million-cycle simulator campaigns; for edge loading and high-flexion pivoting, we use targeted benchtop and component-level tests. For backside micromotion, we rely on insert–tray shear rigs with measured nodal displacements. Scores are assigned per pair according to the number of outputs compared, the statistical agreement, and the independence of the tests.
Output comparison — gradation.

a.
No direct comparisons of model outputs to measured quantities.

b.
Qualitative or single-output comparisons with limited statistical assessment.

c.
Multiple quantitative comparisons with error metrics and independent data sources.

d.
Extensive, mechanism-specific comparisons across outputs with statistical bounds and replication.

14. Credibility Scoring by Model–Mechanism Pair: Results Robustness

Robustness reflects stability of conclusions under plausible alternative inputs and formulations. We tested friction models, contact enforcement stiffness levels, mesh patterns, and

wear coefficient variants within documented ranges. We also compared kinematics-driven and load-driven instantiations for overlapping mechanisms to assess whether rankings of risky conditions persist. Per-pair scores reflect the breadth and impact of these stress tests.
Results robustness — gradation.

a.
No sensitivity or alternative-formulation checks.

b.
Limited sensitivities show changes but without assessment of impact on conclusions.

c.
Systematic sensitivities and select alternative formulations demonstrate stable conclusions.

d.
Extensive stress testing across multiple plausible alternatives confirms conclusions are insensitive within decision tolerances.

15. Credibility Scoring by Model–Mechanism Pair: Results Uncertainty

We propagated uncertainty in friction, cross-shear exponents, wear coefficients, and ligament slack lengths using Latin hypercube sampling and computed Sobol indices to apportion variance. We report predictive intervals for wear volumes and micro-slip and use these to gauge decision margins. Scores account for the comprehensiveness of parameters varied and the clarity of interval reporting.
Results uncertainty — gradation.

a.
No uncertainty quantification.

b.
Qualitative uncertainty discussion without quantitative propagation.

c.
Quantitative propagation across key parameters with reported intervals.

d.
Comprehensive uncertainty quantification with global sensitivity, validated distributions, and decision-focused interval reporting.

16. Credibility Scoring by Model–Mechanism Pair: Test Conditions

External laboratory protocols vary in lubricant properties, waveform phasing, and component alignment. Where necessary, we harmonized inputs or documented deviations, and we report scores reflecting the proximity of compared tests to the modeled conditions while noting that not all apparatus details could be matched.
Test conditions — gradation.

a.
No alignment between modeled scenarios and any test conditions.

b.
Partial alignment; key differences in apparatus or waveform are unaccounted for.

c.
Good alignment with documented mapping and minor residual differences.

d.
Near-exact alignment with well-documented mapping of all key apparatus and waveform elements.

17. Credibility Scoring by Model–Mechanism Pair: Use Error

We considered potential misconfigurations such as incorrect ligament slack lengths, sign errors in tibial slope, or inadvertent changes to friction parameters. We introduced guardrails in pre-processing templates to minimize such risks and included checks of expected ranges for contact and energy metrics. Scores reflect the presence of these safeguards and the residual risk of misuse.
Use error — gradation.

a.
High likelihood of misuse with no safeguards.

b.
Some documentation exists, but misconfiguration remains likely and impactful.

c.
Templates, checks, and documentation reduce likelihood; impact is bounded.

d.
Strong safeguards, automated checks, and training materials make misuse unlikely and quickly detectable.

18. Credibility Scoring by Model–Mechanism Pair: Use History

The models have been exercised in internal studies and limited external collaborations for design comparisons and what-if analyses. Broader deployment histories across institutions and long-term usage patterns are still emerging; we reflect the current breadth and maturity in the assigned indices without asserting generalized adoption.
Use history — gradation.

a.
No prior use in relevant problems.

b.
Limited internal use with minimal external exposure.

c.
Documented use in multiple relevant studies with some external users.

d.
Established use across organizations and decision contexts with documented outcomes.

19. Credibility Scoring by Model–Mechanism Pair: Numerical Solver Error

For completeness of presentation, we reiterate that step-size and mesh refinement checks were performed under each mechanism to ensure stable solutions. The supplemental matrices capture the specific indicators collected and the resulting indices.

Numerical solver error — gradation.

a.
No numerical quality evidence.

b.
Spot checks only; demanding cases not addressed.

c.
Refinement and stability checks for relevant cases with small changes upon tightening.

d.
Thorough convergence demonstrations across all cases with predefined tolerances.

20. Synthesis of Findings and Credibility Elevation Priorities

Across mechanisms, the component-level and anatomy-coupled models provided consistent pictures of contact and wear behaviors, with load-driven ligament constraints moderating extremes in edge loading and pivoting at lower flexion angles. The supporting evidence for parameterization and comparison is strong for level walking wear and contact patch localization, moderate for edge loading and high-flexion pivoting, and emergent for backside micromotion under tolerance excursions. Given the mapping between measured shear rig outputs and the computational micro-slip, the interface behavior is credible under nominal assembly but becomes sensitive to assembly or tolerance variations that are not always captured in standard test protocols.

We observed that the set of comparisons against physical tests for gait resulted in a root-mean-square discrepancy of 12 percent in volumetric wear after calibrating a single coefficient, while the scar localization aligned at a Jaccard index above 0.72, forming the basis for trusting the model for relative design comparisons in that regime. For numerical quality, constraining the explicit integration to maintain a minimum time step of 0.12 microseconds kept kinetic-to-internal energy ratios below 2 percent and hourglass energy below 0.5 percent; halving the time step altered peak contact pressures by less than 0.8 percent, supporting the stability of incremental wear calculations.

The sources underpinning parameters and comparisons are diverse and well documented; three independent GLP campaigns, specimen-linked UHMWPE properties, and locking feature measurements created a coherent chain from physical artifacts to model inputs and outputs. We selected the constitutive structure purposefully to emphasize deformation and sliding relevant to wear while excluding poroviscoelastic diffusion and oxidative chemistry; adopting a Kelvin–Voigt element for creep reproduced residual thickness trends without invoking more elaborate viscoplastic frameworks.

We found that alternatives in friction formulation and mesh topology changed volumetric wear by less than 9 percent across mechanisms and did not alter the ranking of risky conditions, indicating that conclusions about which mechanisms drive higher wear are stable to plausible modeling choices. Uncertainty quantification illuminated the drivers of spread: Sobol analysis showed that friction and the cross-shear exponent account for 72 percent of variance in wear under high-flexion pivoting, while condylar radius tolerances contributed under 5 percent; the 95 percent predictive interval for 1 million-cycle wear volume under gait spans 24 percent, providing usable margins for comparative decisions.

Finally, when we intentionally mis-specified ligament slack lengths by 2 mm or reversed the sign of tibial slope in pre-processing, the anatomy-coupled model produced unstable roll-back and rim engagement that violated workflow expectations; in response, we locked those inputs behind templates and added range checks to flag out-of-family configurations.

Looking forward, the most effective steps to elevate credibility scores into the established tier for the challenging mechanisms are: (1) dedicated edge-loading experiments mapping angulation and lift-off to contact and wear outputs, enabling direct apples-to-apples comparisons with the modeled malalignments; (2) high-flexion pivoting wear tests that report cross-shear metrics alongside volumetric loss to sharpen calibration beyond gait; (3) systematic characterization of insert–tray locking compliance across sizes and tolerances, including post-assembly gap evolution; and (4) expanded uncertainty quantification with validated distributions for friction and lubrication states across activities, thereby tightening predictive intervals.

21. Limitations

The explicit FEA models emphasize contact mechanics and wear accumulation but do not include polyethylene oxidation kinetics, third-body abrasion, or fluid film lubrication with Reynolds solvers. While the chosen friction models and cross-shear scaling are grounded in experiments, they are simplified representations of tribological behavior. The titanium tray is treated as rigid in Model 1 and as a thin elastic foundation in Model 2; more detailed tray compliance might modestly alter backside micromotion distributions. The ligament models are spring surrogates that capture gross laxity envelopes but not fiber-level wrapping or rate dependence. Moreover, the high-flexion pivoting and edge-loading scenarios are mapped from benchtop surrogates rather than full-joint activities of daily living, and external apparatus differences in lubricant composition and waveform phasing are not exhaustively reconciled. Finally, while internal and limited external uses provide encouraging signals, long-term, multi-institutional usage patterns are not yet in place for these specific models and workflows.

22. Conclusions

Two explicit FEA models of a posterior-stabilized total knee replacement were evaluated across four clinically relevant mechanisms for contact, wear, and interface behavior. The component-level, kinematics-driven model reproduced gait wear volumes and wear scar distributions within 12 percent RMS after calibrating a single coefficient and captured the expected sensitivity of contact pressure and wear to edge loading and pivoting. The anatomy-coupled, load-driven model added ligament restraint that moderated extremes in rim contact and high-flexion pivoting without changing the qualitative rankings of risky conditions. Backside interface micromotion remained below 15 microns under nominal locking and increased substantially when assembly or tolerance excursions were imposed. A structured credibility assessment using a 0–12 index demonstrated developing-to-established support for gait wear predictions and developing support for edge loading, backside micromotion, and high-flexion pivoting. Priority evidence needs include direct edge-loading and pivoting wear tests with cross-shear reporting and expanded characterization of insert–tray compliance. Within these bounds, the models support comparative design assessment and what-if studies, with caution for extreme postures and interface tolerances.

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Table 1: Summary of cross-cutting findings and quantitative anchors used in credibility interpretation. Each row restates a statement made in the body prose and provides the specific numeric values.

Credibility factor	Level	Basis
Evidence traceability	Global	Three independent GLP simulator campaigns with specimen-linked UHMWPE properties and locking feature tests underpinned parameters and comparisons.
Governing choices	Global	A Kelvin–Voigt creep element combined with cross-shear–dependent Archard wear reproduced residual thickness trends without invoking poro-viscoelastic diffusion.
Implementation checks	Global	Thirty-eight unit tests plus two manufactured-solution cases achieved L2 errors below 1e-6 for displacement and pressure fields.
Integration quality	Global	Minimum time step 0.12 microseconds kept kinetic-to-internal energy ratios under 2 percent and hourglass energy under 0.5 percent; halving the step changed peak pressure by less than 0.8 percent.
Agreement with experiments	Gait	After calibrating one global wear coefficient, RMS error in volumetric wear was 12 percent and wear scar Jaccard index exceeded 0.72.
Stability to alternatives	Global	Switching friction models (0.02–0.08 Coulomb vs velocity-weakening) and meshing strategies shifted wear volume by at most 9 percent without changing mechanism rankings.
Uncertainty characterization	Global	Sobol analysis attributed 72 percent of wear variance under high-flexion pivoting to friction and cross-shear exponent; the 95 percent interval for 1 Mc wear volume spans 24 percent.
Safeguards against misuse	Global	Mis-specified ligament slack by 2 mm or reversed tibial slope produced unstable rollback; templates and range checks were added to prevent and flag such inputs.